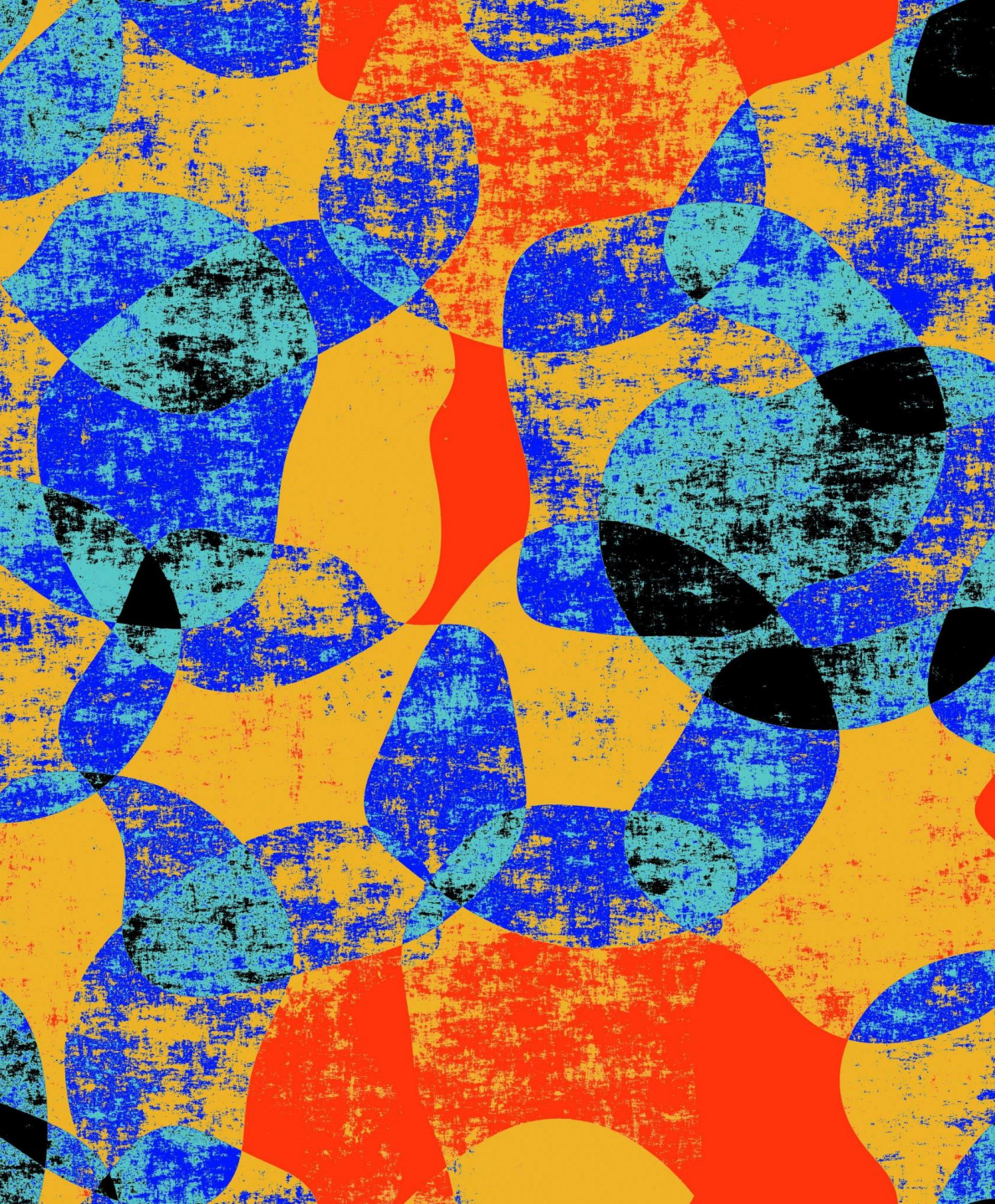




---

## ASSIGNMENT 5: COMMUNICATION: ENABLING SENTIMENT ANALYSIS

- Group 03
- Kamil Bienias
- Hongqian Li
- Marko Paradzik
- Muslim Ramazanov
- Chamberg Zelimchanow

# app-clco-demo-group-3

Web App

- Browse
- Stop
- Swap
- Restart
- Delete
- Refresh
- Download publish profile
- Reset publish profile
- Share to mobile
- Send us your feedback

## Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Microsoft Defender for Cloud
- Events (preview)
- Log stream
- Resource visualizer
- Deployment
  - Deployment slots
  - Deployment Center
- Settings
- Performance
- App Service plan
- Development Tools
- API
- Monitoring
- Automation

Consider using Azure Front Door since your app receives incoming requests from multiple regions.

### Essentials

|                                         |                                        |                  |                                                            |
|-----------------------------------------|----------------------------------------|------------------|------------------------------------------------------------|
| Resource group ( <a href="#">move</a> ) | : <a href="#">rg-clco-demo-group-3</a> | Default domain   | : <a href="#">app-clco-demo-group-3.azurewebsites.net</a>  |
| Status                                  | : Running                              | App Service Plan | : <a href="#">asp-clco-demo-group-3 (B1: 1)</a>            |
| Location ( <a href="#">move</a> )       | : Norway East                          | Operating System | : Linux                                                    |
| Subscription ( <a href="#">move</a> )   | : <a href="#">Azure for Students</a>   | Health Check     | : <a href="#">Not Configured</a>                           |
| Subscription ID                         | : 49b9ff2a-051e-43f3-8aea-1cb023d131c8 | GitHub Project   | : <a href="#">https://github.com/hongqian-li/clco-demo</a> |
| Tags ( <a href="#">edit</a> )           | : <a href="#">Add tags</a>             |                  |                                                            |

- Properties
- Monitoring
- Logs
- Capabilities
- Notifications (1)
- Recommendations

### Web app

|                  |                       |
|------------------|-----------------------|
| Name             | app-clco-demo-group-3 |
| Publishing model | Code                  |
| Runtime Stack    | Python - 3.11         |

### Deployment Center

|                     |                                                                             |
|---------------------|-----------------------------------------------------------------------------|
| Deployment logs     | <a href="#">View logs</a>                                                   |
| Last deployment     | Successful on Tuesday, November 12, 2024 at 1:00 PM <a href="#">Refresh</a> |
| Deployment provider | GitHubAction                                                                |

### Domains

|                |                                                         |
|----------------|---------------------------------------------------------|
| Default domain | <a href="#">app-clco-demo-group-3.azurewebsites.net</a> |
| Custom domain  | <a href="#">Add custom domain</a>                       |

### Application Insights

|      |                                             |
|------|---------------------------------------------|
| Name | <a href="#">Enable Application Insights</a> |
|------|---------------------------------------------|

### Networking

# WEB APP OVERVIEW







## Cloud Computing Project

Deployed with Azure Terraform & GitHub Actions

Team: Kamil, Hongqian, Marko, Muslim, Chamberg

Tell me your name and message!

Submit

We had the following visitors

Mood of the last ten visitors [Analyse sentiment](#)

| Name        | Message                                      | Visited                    |
|-------------|----------------------------------------------|----------------------------|
| Tom         | Hello, world!                                | 2022-06-07 20:31:17        |
| Peter       | I like Pizza                                 | 2022-07-17 03:00:38        |
| Max         | carpe diem                                   | 2022-10-28 15:01:17        |
| Robert      | woof                                         | 2022-05-18 15:01:51        |
| Michael     | This is cool                                 | 2023-08-24 17:19:00.236157 |
| Peter       | this is just awful                           | 2023-08-24 17:19:14.273265 |
| Max         | I just wish I was home                       | 2023-08-24 17:19:23.730193 |
| Vishnu      | Now I have become death, destroyer of worlds | 2023-08-24 17:19:43.174624 |
| Hongqian Li | Hello!                                       | 2025-11-18 19:21:48.630375 |
| Kamil       | This is really bad!                          | 2025-11-18 19:25:35.983503 |

OUR APP IS  
WORKING!

---

portal.azure.com/#@technikum-wien.at/resource/subscriptions/49b9ff2a-051e-43f3-8aea-1cb023d131c8/resourceGroups/rg-clco-demo-group-3/providers/Microsoft.C...

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Home > Azure for Students | Resource groups > rg-clco-demo-group-3 > ca-clco-demo-group-3

ca-clco-demo-group-3 | Networking ☆ ...

Language

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Resource Management

Features

Keys and Endpoint

Encryption

Commitment tier pricing

Pricing tier

Networking

Identity

Cost analysis

Properties

Security

Monitoring

Automation

...

Firewalls and virtual networks Private endpoint connections

Save Discard Refresh

Allow access from

☐ All networks ☐ Selected Networks and Private Endpoints ☒ Disabled

No networks can access this resource. Private endpoint connections will be the exclusive way to access this resource. [Learn more.](#)

Exceptions

☒ Allow Azure services on the trusted services list to access this cognitive services account. ⓘ

# LANGUAGE SERVICE NETWORKING CONFIGURATION (1)

---

portal.azure.com/#@technikum-wien.at/resource/subscriptions/49b9ff2a-051e-43f3-8aea-1cb023d131c8/resourceGroups/rg-clco-demo-group-3/providers/Microsoft.C...

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home > Azure for Students | Resource groups > rg-clco-demo-group-3 > ca-clco-demo-group-3

ca-clco-demo-group-3 | Networking ☆ ...

Language

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Resource Management

Features

Keys and Endpoint

Encryption

Commitment tier pricing

Pricing tier

Networking

Identity

Cost analysis

Properties

> Security

> Monitoring

> Automation

> Help

Firewalls and virtual networks

Private endpoint connections

+ Private endpoint ✓ Approve ✕ Reject 🗑 Remove ↻ Refresh

Filter by name...

All connection states

| <input type="checkbox"/> Connection name                         | Connection state | Private endpoint        | Description |
|------------------------------------------------------------------|------------------|-------------------------|-------------|
| <input type="checkbox"/> ca-clco-demo-group-3-pe.ffb3dbf1-6d6... | Approved         | ca-clco-demo-group-3-pe | Approved    |

# LANGUAGE SERVICE NETWORKING CONFIGURATION (2)

---

rg-clco-demo-group-3

Resource group

Export resource groups using Bicep or Terraform

+2

✕

Search

◇

◀

+

Create

⚙️

Manage view

🗑️

Delete resource group

🔄

Refresh

⋮

☰

Group by none

▼

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Essentials

JSON View

Subscription (move)

[Azure for Students](#)

Subscription ID

49b9ff2a-051e-43f3-8aea-1cb023d131c8

Deployments

No deployments

Location

Norway East

Tags (edit)

[Add tags](#)

Resources

Recommendations

Filter for any field...

Type equals all

Location equals all

+ Add filter

| <input type="checkbox"/> | Name ↑                                                    |   | Type              | Location    |
|--------------------------|-----------------------------------------------------------|---|-------------------|-------------|
| <input type="checkbox"/> | <a href="#">app-clco-demo-group-3</a>                     | ⋮ | App Service       | Norway East |
| <input type="checkbox"/> | <a href="#">asp-clco-demo-group-3</a>                     | ⋮ | App Service plan  | Norway East |
| <input type="checkbox"/> | <a href="#">ca-clco-demo-group-3</a>                      | ⋮ | Language          | Norway East |
| <input type="checkbox"/> | <a href="#">ca-clco-demo-group-3-pe</a>                   | ⋮ | Private endpoint  | Norway East |
| <input type="checkbox"/> | <a href="#">ca-clco-demo-group-3-pe.nic.1a5c30ac-df32</a> |   | Network Interface | Norway East |
| <input type="checkbox"/> | <a href="#">oidc-msi-ae87</a>                             | ⋮ | Managed Identity  | Norway East |
| <input type="checkbox"/> | <a href="#">privatelink.cognitiveservices.azure.com</a>   | ⋮ | Private DNS zone  | Global      |
| <input type="checkbox"/> | <a href="#">vn-clco-demo-group-3</a>                      | ⋮ | Virtual network   | Norway East |

Showing 1 - 8 of 8. Display count:

auto

▼

Give feedback

# TERRAFORM- DEPLOYED RESOURCES

# SENTIMENT ANALYSIS BUTTON IN THE WEB APP

---

app-clco-demo-group-3.azurewebsites.net

 **CLCO**

 **Cloud Computing Project**  
Deployed with Azure Terraform & GitHub Actions  
Team: Kamil, Hongqian, Marko, Muslim, Chamberg

**Sentiment analysis**

|                                                     |                                     |
|-----------------------------------------------------|-------------------------------------|
| <i>Hello, world!</i>                                | <b>Neutral with 0.7 certainty</b>   |
| <i>I like Pizza</i>                                 | <b>Positive with 1.0 certainty</b>  |
| <i>carpe diem</i>                                   | <b>Neutral with 0.94 certainty</b>  |
| <i>woof</i>                                         | <b>Neutral with 0.59 certainty</b>  |
| <i>This is cool</i>                                 | <b>Positive with 0.98 certainty</b> |
| <i>this is just awful</i>                           | <b>Negative with 1.0 certainty</b>  |
| <i>I just wish I was home</i>                       | <b>Positive with 0.72 certainty</b> |
| <i>Now I have become death, destroyer of worlds</i> | <b>Negative with 0.96 certainty</b> |
| <i>Hello!</i>                                       | <b>Positive with 0.85 certainty</b> |
| <i>This is really bad!</i>                          | <b>Negative with 1.0 certainty</b>  |

Close

| Name        | Message                                      | Timestamp                  |
|-------------|----------------------------------------------|----------------------------|
| Tom         |                                              |                            |
| Peter       | I like Pizza                                 | 2022-07-17 03:00:38        |
| Max         | carpe diem                                   | 2022-10-28 15:01:17        |
| Robert      | woof                                         | 2022-05-18 15:01:51        |
| Michael     | This is cool                                 | 2023-08-24 17:19:00.236157 |
| Peter       | this is just awful                           | 2023-08-24 17:19:14.273265 |
| Max         | I just wish I was home                       | 2023-08-24 17:19:23.730193 |
| Vishnu      | Now I have become death, destroyer of worlds | 2023-08-24 17:19:43.174624 |
| Hongqian Li | Hello!                                       | 2025-11-18 19:21:48.630375 |
| Kamil       | This is really bad!                          | 2025-11-18 19:25:35.983503 |

---

# SUMMARY

- We completed the assignment by deploying our web app and connecting it with Azure Cognitive Services for sentiment analysis. We faced challenges with the private endpoint setup, especially needing to add a custom subdomain name and link the DNS zone correctly. We also had to learn how to integrate the web app with the virtual network using the Swift connection. With AI assistance, we quickly identified these issues and fixed our Terraform code. Overall, we gained experience in secure networking and automation, and we successfully got sentiment analysis working in our app.

